HOSPITALITY SUSTAINABILITY AI

A transparent, explainable machine-learning prototype for hospitality electricity prediction and decision support

RESEARCH PORTFOLIO REPORT

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Project focus	Explainable AI, energy modelling, anomaly investigation and responsible human oversight
Repository	github.com/akshith2001/hospitality-sustainability-ai
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Research position

This project is an educational research prototype, not a deployed energy-control system. Its purpose is to demonstrate a reproducible and honest workflow connecting prediction, evaluation, explanation, anomaly investigation, intervention comparison and human approval.

1. Abstract and research contribution

Hospitality venues face highly variable electricity demand shaped by weather, season, building characteristics and operations. This project investigates whether a transparent machine-learning model can improve short-term prediction while preserving interpretability, data-quality controls and human decision authority. The repository combines synthetic learning experiments with a real-building validation using Building Data Genome 2 (BDG2).

The real-data experiment used 5,755 daily records from six food-service buildings and two hotels. A chronological design trained on eligible historical observations through 1 November 2017 and evaluated the newest 60 dates. Against a per-venue historical-mean baseline, the seasonal linear model reduced overall mean absolute error (MAE) from 651.73 to 523.42 kWh/day, an improvement of 19.7%. The model improved performance for four of five eligible test venues and performed worse for one. Three source venues had no eligible test-period readings and were explicitly reported as missing.

5,755	8	2	92
REAL DAILY RECORDS	SOURCE BUILDINGS	UNSEEN VENUES TESTED	AUTOMATED TESTS

Key contributions

- A reproducible pipeline that converts hourly electricity and weather data into quality-controlled daily records.
- A transparent seasonal ordinary least-squares model evaluated without future-information leakage.
- Venue-level reporting that includes weaker, missing and incomplete outcomes rather than selecting favourable cases.
- An explainability and governance design that treats alerts as evidence for investigation, not proof of waste.
- A proposed real-world study protocol using verified meter readings, pseudonymous identifiers and human approval gates.

Research question

Can a transparent seasonal linear model predict the newest 60 days of daily electricity use more accurately than a per-venue historical-mean baseline for hospitality-related buildings selected from BDG2?

2. Data and methodology

Data source

The real-data study uses Building Data Genome Project 2, version 1.0. The importer selects six buildings classified as Food sales and service and two lodging buildings whose subindustry is Hotel. Hourly electricity readings are aggregated to daily totals when at least 20 hourly values are present, and daily mean outdoor temperature is joined by site and date. The resulting subset covers 1 January 2016 to 31 December 2017.

Evaluation design

Element	Implementation
Training boundary	All eligible observations up to and including 1 November 2017
Held-out period	Newest 60 dates, beginning 2 November 2017
Baseline	Each venue's mean daily kWh calculated from training data only
Model	Ordinary least squares using venue identity, outdoor-temperature terms, weekday, annual seasonality and venue-specific annual seasonality
Primary metric	Mean absolute error in kWh/day; lower is better
Quality rule	Daily totals below 1 kWh excluded as zero/near-zero meter states for buildings above 1,500 square metres; applied before splitting

Data-quality disclosure

The near-zero rule was introduced after the first diagnostic run exposed prolonged zero or fixed tiny readings. It is therefore a post-diagnostic quality decision, not a pre-registered choice. This is disclosed to prevent the improved aggregate result from being presented as if every decision had been fixed in advance.

Software verification

The repository contains 88 automated tests covering modelling, baselines, temporal and venue validation, uncertainty, anomaly detection, explainability, recommendations, governance, meter quality, operational context, real-data preparation and diagnostics. The full suite was rerun successfully for this report on 16 August 2026.

3. Results and interpretation

Overall test performance

Evaluation	Test rows	MAE (kWh/day)	Result
Per-venue historical mean	266	651.73	Baseline
Seasonal linear model	266	523.42	19.7% lower MAE

Venue-level outcome

• Five venues had eligible observations in the held-out period.

• The model beat the baseline for four venues and underperformed for Fox_food_Scott.

• Eagle_food_Jennifer contributed 26 eligible test days rather than the full 60.

• Lamb_food_Sylvia, Lamb_lodging_Harley and Mouse_lodging_Vicente had no eligible test-period readings and were reported as missing.

Interpretation

The result provides evidence that weather, calendar and seasonal features add predictive value beyond a simple historical mean for most eligible venues in this sample. It also shows that performance is uneven. The weaker venue and the missing test periods are substantive findings because they expose limits that an overall average could conceal.

What the result does not prove

• It does not establish causality or prove that high consumption represents waste.

• It does not verify intervention savings or operational behaviour changes.

• It does not validate the full hospitality feature set because BDG2 lacks customer counts, opening hours and kitchen-equipment counts.

• The chronological experiment alone does not establish transfer to unseen venues because that model uses venue identity.

The appropriate conclusion is therefore modest: the prototype demonstrates promising, reproducible real-data predictive value while clearly identifying where evidence is missing or weaker.

Completely unseen-venue experiment

A separate experiment removed venue identity and held out one food-service building and one hotel completely. The venues were locked before scoring using record count and alphabetical tie-breaking. Success required the model to beat a same-type training mean for both venues separately.

Held-out venue	Baseline MAE	Model MAE	Improvement
Fox_food_Francesco	726.13	327.85	54.8%
Mouse_lodging_Vicente	1,001.15	676.23	32.5%

The predefined criterion was met. However, this is a cross-building transfer test, not future-time forecasting: the training venues include observations from the same date range. Two buildings are insufficient to establish broad generalisation.

4. Responsible AI design and next research steps

Explainability and decision support

For the project's interpretable linear models, each feature contribution is calculated as the input value multiplied by its learned coefficient, and contributions sum to the prediction. These values explain the model calculation but are not presented as causal effects. Residual-based alerts request investigation; legitimate events, weather, equipment faults and missing variables may all produce unusual readings.

Governance safeguards

- Recommendations require authorised human approval and strong alternatives are shown.
- Performance is reported overall and by venue to expose unequal outcomes.
- Material degradation pauses recommendations and triggers human review.
- Personal names, business names and full addresses are excluded from training files.
- A real-world study would require institutional ethics and data-protection approval.

Priority next experiments

Priority	Experiment	Why it matters
1	Compare stronger time-series baselines	Tests whether the seasonal model improves on persistence, rolling means and seasonal naive forecasts
2	Combine unseen venue and future holdout	Tests building and temporal transfer simultaneously
3	Pre-register quality and evaluation rules	Reduces the risk of post-diagnostic choices influencing reported performance
4	Collect consented operational context	Adds occupancy, opening hours and equipment information unavailable in BDG2
5	Evaluate intervention outcomes	Separates prediction accuracy from verified energy and emissions savings

Conclusion

Hospitality Sustainability AI is a credible early-stage research portfolio project because it combines working software with transparent assumptions, chronological evaluation, venue-level reporting, automated tests and explicit limitations. Its strongest contribution is not a claim of proven savings; it is a disciplined framework for learning from imperfect building data while keeping recommendations explainable and under human control.

References and reproducibility

Primary data

Miller, C., Kathirgamanathan, A., Picchetti, B., Arjunan, P., Park, J. Y., Nagy, Z., Raftery, P., Hobson, B. W., Shi, Z., and Meggers, F. Building Data Genome Project 2, version 1.0. Zenodo. DOI: 10.5281/zenodo.3887306. Licensed CC BY 4.0.

Project materials

Moharampudi, A. Hospitality Sustainability AI, version 0.1.0. MIT License. Repository: https://github.com/akshith2001/hospitality-sustainability-ai

The repository contains the importer, derived daily subset, evaluation code, tests, model card, data-source record, real-world study protocol and full real-data result note used to prepare this report.

Reproducibility summary

- Python 3.10 or newer; project code uses the Python standard library.
- BDG2 source ZIP integrity verified against the official MD5 checksum.
- Daily aggregation requires at least 20 hourly readings.
- Chronological train/test boundary and near-zero exclusion rule are documented.
- All 92 automated tests passed during report preparation.

Contact and portfolio

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Project: github.com/akshith2001/hospitality-sustainability-ai